textarea

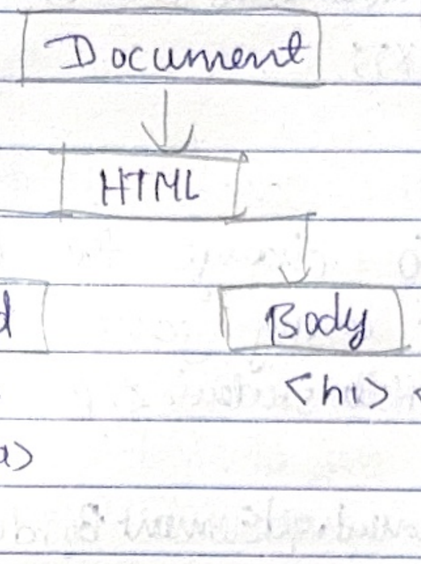
It is the element used to except multi line text input. from the user.

placeholder

It displays the inte before typing

DOM - Document object model

DOM - It is a programming interface that represents html object as ~~tree~~ of object



DOM is used to access, modify, add or delete html element, attributes, style in java script.

Objects of DOM

1) Document

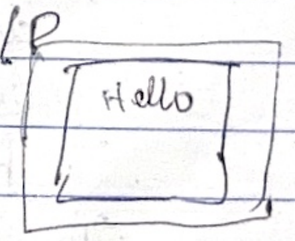
It represents entire/whole html file

2) window

It is a object that represents browser window

Ex:- <body>
<script>
window.alert ("Hello")
</script>
</body>

O/P



3) element:

An element object represent individual html element such as <p> <h1> and <button> etc.

Ex:- <h1 id = "title" > Hellow </h1>
<script>
let x = document.getElement("title");
document.write (x);
</script>

4) style: It is used to change the css styling of an html element using java script

Ex:- <P id = "demo" > Hello students </P>
<script>
document.write (document.getElementById("demo").style.color="red");
</script>

5) class name

This property is used to get or change the css class name of an html element.

Ex:- <!doctype html>
<html>
<head>
<title> DOM class </title>
<style>
.highlight {
color : red;
background-color : yellow;
font-size : 25px;
}
</style>
<body>
<P id = "demo" > Hello students </P>
<script>
document.write (document.getElementById ("Demo").className = "highlight");

hello student

</script>
</body>
</html>

6) classlist

It is a DOM object used to add, remove, toggle and check css class of in html using java script.

method

add() -> adds a css class to an element.
remove() -> removes a css class from an element.
toggle() -> adds class if absent or removes class if present.
contains() -> Check whether element contains a class

add()

Ex:- <!doctype html>
<html>
<head>
<style>
.highlight {
color : red;
background-color : yellow;
}
</style>
</head>
<body>
<P id = "demo" > Hello student </P>
<script>
document.write (document.getElementBy Id ("demo").
className = "highlight");
document.write (document.getElementBy("Demo").classList.remove
("highlight");
</script>
</body>
</html>

7) queary Selector()
8) innerHTML()
9) TextContent()
10) innerText()

innerHTML()

It is a property of DOM used to get or set the html content of an html element.

① get innerHTML

It retrives existing html of an element

Ex:- hi

```
<script>
document.write(document.getElementById("demo").innerHTML);
</script>
</body>
```

② set innerHTML

It changes or relplaces existing html element content.

Ex:- good morning

```
<script>
document.getElementById("demo").innerHTML = "Hello Student";
</script>
</body>
```

innerText()

It is the property of DOM used to get or change visible text of an html element

Note:

It doesinot retrive html tags

Ex:- <body>

```
<p id="Demo"><b> hi </b></p>
<script>
document.write(document.getElementById("demo").innerText);
</script>
</body>
```

Set innerText()

Ex:- <body>

```
<p id="demo"><u> hi </u></p>
<script>
document.getElementById("demo").innerText = "Hello";
</script>
</body>
```

TextContent()

It is a DOM property used to get or set textcontent of html element.

→ It does not interprit html tags :- Note

Hidden content also be display using textcontent

Ex:- <body>

```
<p id="demo">Hello</p>
<p id="some" style="display: none">some secret
    manage</p>
<script>
let x = document.getElementById("demo").innerHTML;
document.write(x);
let y = document.getElementById("some").textContent;
document.write(y);
</script>
</body>
```

CSS in DOM

It is used to style the DOM document.

Syntax:-

document.getElementById("idname").style.attribute;

Ex:-
```
<body>
<p id="demo"> Hello </p>
<script>
document.getElementById("demo").style.color = "Red";
    • fontSize = 25px;
    • backgroundcolor = Yellow;
    • border = 2px solid green;
</script>
</body>
```

Document.querySelector

It is a DOM property used to select first HTML element that matches specified CSS selector.

Ex:-
```
<html>
<body>
<p> Hello student </p>
<p> welcome to BCA </p>
<script>
letx = document.querySelector("p").style.color = "red";
document.write(x);
</script>
</body>
</html>
```

Form validation

Process of checking whether user as entered or selected required information currectly before to submiting a form.

Ex: program 20

DOM Navigation

It is a process of moving b/w html element in a document object model using java script

It allows us to access parent, child and sibling element of an html document.

Ex:-
```
<!doctype html>
<html>
<body>
<div id = "parent">
<p id = "first"> first paragraph </p>
<p id = "second" > second paragraph </p>
</div>
<script>
let second = document.getElementById ("second");
console.log (second.parentElement);
console.log (second, previousElement sibling);
console.log (second.parentElement.firstElementchild);
</script>
</body>
</html>
```

O/p:
```
<div id = "parent">
<p id = "first"> first paragraph </p>
<p id = "first"> first paragraph </p>
```

```
<!doctype html>
<html>
<body>
<div id = "college">
<h2> welcome to BCA </h2>
<p id = "msg"> Hello student! </p>
</div>
<script>
let msg = getElementById ("msg");
console.log (msg. parentElement);
console.log (msg. previousElement sibling);
console.log (msg. parentElement. children);
</script>
</body>
</html>
```

o/p =
<h2> welcome to BCA </h2>
<h2> welcome to BCA </h2>
 Hello students!

Note:

- parentElement - parent
- children - all children
- firstElementchild - It returns first child element of parent
- lastElementchild - It returns last child element of parent
- previousElementsibling - It returns the previous sibling Element
- nextElementsibling - It returns the next sibling element

DOM events

Events are the actions or the occurrences that happen in a web browser and can be detected by javascript

They allow webpage to respond to user interaction such as clicking a button, pressing a key, moving a mouse or submitting a form.

Types of DOM events.

Events	Description
onclick	occurs when a element is clicked
ondblclick	occurs when a element is double clicked
onmouseover	occurs when a mouse pointer move over a element
onkeydown	occurs when key is pressed down
oninput	occurs when value of a input changes
onsubmit	occurs when the form is submitted.

Event listener

Event listener is used to execute a function when a specific event occur on html element

addEventListener() is the method use to attach event listeners.

Syntax:

element. addEventListener ("event", function);

Ex:- <button id = "btn"> click me </button>

```
<script>
document.getElementById ("btn"). addEventListener ("click",
```

```
function (){
    alert ("Button clicked")
    });
</script>
```

O/P = [click me]

Button clicked

```
<!doctype html>
<html>
<body>
<h2 id = "demo"> move mouse over me </h2>
<script>
document.getElementById ("demo").addEventListener ("mous
    -over", function(){
document.getElementById ("demo").innerHTML = "mouse
    is over text!"});
</script>
</body>
</html>
```

O/P = move mouse over me.

DOM node

DOM node is a individual part of an html document

Every element, attribute, text and comment in an html document is represented as node in DOM tree

Type of DOM nodes

① Element node - It represents the html elements or tags Ex: <h2>

② Text node: ~~represents the html elements~~ represents the HTML text inside the html element

Ex: hello

text node = Hello

3. attribute node - It represents the html attributes such as ID, class or href

Eg: <h2 id = "demo"> move mouse over me </h2>

attribute node = id

4. comment node: It represents the comments written in the html document

Eg: <!-- Example for Events -->

node type	value
Element node	1
Attribute node	2
text node	3
comment node	8
documenttype node	10

Ex-

```
<!doctype html>
<html>
<body>
<p id = "demo"> Hello </p>
<script>
let p = document.getElementById ("demo");
console.log (p.nodetype);
console.log (p.nodeName);
console.log (p.textcontent);
```

</script>
</body>
</html>

O/P - 1
P
Hello

```
<!doctype html>
<html>
<body>
<p id="demo"> Hello </p>
<script>
let p=document.getElementById("demo");
console.log (p.nodeType);
console.log (p.~~node~~ document, nodeType);
</script>
</body>
</html>
```

O/P - 1
10

DOM node list - Document object model node list is the collection of DOM nodes selected ~~from~~ from html document.
It allows us to access multiple nodes together

syntax
document.queryselectorAll("element");

Ex:-

```
<!doctype html>
<html>
<body>
<p> First paragraph </p>
<p> Second paragraph </p>
<p> Third paragraph </p>
<script>
let nodes = document.queryselectorAll("p");
console.log (nodes);
console.log (nodes.length);
console.log (nodes[0].textcontent);
</script>
</body>
</html>
```

O/P - P, P, P
3
First paragraph

DOM node collection
DOM node collection is a group of DOM nodes returned by certain javascript methods
There are two type of collections;

1. HTML collection
2. node list

1 HTML collection - It is a collection of HTML elements returns by certain DOM methods in JS

syntax:
document.getElementByTagName("tag")

Ex:- <!doctype html>
<html>

```
<body>
 <P> First paragraph </P>
 <P> second paragraph </P>
 <P> Third paragraph </P>
 <script>
 let elements = document.getElemenntBy TagName("P")
  console.log (elements);
  console.log (elements.length);
  console.log (Element[0].textcontent);
  </script>
  </body>
  </html>
o/P: P, P, P
     3
```

ECMA script

ECMA script is a standard specification that defines rules, syntax and features of javascript programming language.

Need of ECMA script (Advantages)

→ provides standard rules for javascript.
→ Ensures consistent javascript behaviour across different browser
→ Introduces new features and improvements.
→ Support model web application development.
→ Improves code readability and maintainability.

EMA features!

→ Variable declaration - let, var, const
→ classes - supports class based systax for creating objects
→ default parameters - allows default values to be assigned to function parameters.
→ modules - support import and export to organise code into separate files.

JQuery - JQuery is a fast, light-weight feature rich javascript liborary use to simplify html document.

Features of JQuery.

* HTML Manipulation
* CSS manipulation
* Event handling - such as key press, mouse click, mouse over
* It provides cross borowser support
* simple syntax
* It provides / supports Animation

AJAX

AJAX stands for Asynchronous Javascript and xml It is a web development technique use to exchange data with a server and update the parts of the web without reloding entire page.

Features of AJAX!

* Asynchronous commication
* partial page update
* Reduced reloading
* It improves user experience
* Data exchange

- Communication with the server.

React JS

It is a open source Javascript library use to build interactive and dynamic user interfaces especially for single page applications. It was developed by Facebook.

Features of React JS:

- Component based architecture - It divides user interface into small reusable components.
- JSX - It is the technique that allows developers to write html like syntax inside the Javascript
- It provides reusable components.
- Dataflow is generally one way from parent to child
- It provides high performance